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3.5 Summary

‘Real’ neurons are complex biophysical and biochemical entities. Before designing a model it is therefore necessary to develop an intuition for what is important and what can be safely neglected. Synapses are usually modeled as specific ion channels that open for a certain time after presynaptic spike arrival. The geometry of the neuron can play an important role in the integration of incoming signals because the effect of synaptic input on the somatic membrane potential depends on the location of the synapses on the dendritic tree. Though some analytic results can be obtained for *passive* dendrites, it is usually necessary to resort to numerical methods and multi-compartment models in order to account for the complex geometry and presence of active ion channels on neuronal dendrites.

Literature

The book ‘Dendrites’ (501) offers a comprehensive review of the role and importance of dendrites from multiple points of view. An extensive description of cable theory as applied to neuronal dendrites can be found in the collected works of Wilfrid Rall (464). NEURON (90) and GENESIS (63) are important tools to numerically solve the system of differential equations of compartmental neuron models. There are useful repositories of neuronal morphologies (see <http://NeuroMorpho.Org> for instance) and of published models on ModelDB <http://senselab.med.yale.edu/modeldb>. The deep layer cortical neuron discussed in this chapter is described in (208). Potential computational consequences of nonlinear dendrites are described in (337).

Exercises

1. Biophysical synapse model and its relation to other models.

- a) Consider Eq. 3.4 and discuss its relation to Eq. 3.2. Hint: (i) assume that the time course $\gamma(t)$ can be described by a short pulse (duration of 1 ms) and that the unbinding is on a time scale $\beta^{-1} > 10 \text{ ms}$. (ii) Assume that the interval between two presynaptic spike arrivals is much larger than β^{-1} .
- b) Discuss the relation of the depressive synapse model in Eq. 3.8 with the biophysically model in Eq. 3.4. Hint: (i) Assume that the interval between two presynaptic spikes is of the same order than β^{-1} . (ii) In Eq. 3.8 consider a variable $x = P_{\text{rel}}/P_0$.

2. Transmitter-gated ion channel.

- Mark for each of the following statements whether it is correct or wrong: a) AMPA channels are activated by glutamate. b) AMPA channels are activated by AMPA. c) If the AMPA channel is open, AMPA can pass through the channel. d) If the AMPA channel is open, glutamate can pass through the channel. e) If the AMPA channel is open, potassium can pass through the channel.

3. Cable equation.

- a) Show that the passive cable equation for the current is

$$\frac{\partial}{\partial t} i(t, x) = \frac{\partial^2}{\partial x^2} i(t, x) - i(t, x) + \frac{\partial}{\partial x} i_{\text{ext}}(t, x), \quad (3.46)$$

- b) Set the external current to zero and find the mapping to the heat equation

$$\frac{\partial}{\partial t} u(t, x) = \frac{\partial^2}{\partial x^2} u(t, x) \quad (3.47)$$

c) Find the solution to the current equation in a) for the infinite cable receiving a short current pulse at time $t = 0$ and show that the corresponding equation for y satisfies the heat equation in b).

4. Nonleaky Cable.

a) Redo the derivation of the cable equation for the case of an infinite one-dimensional passive dendrite without transversal leak and show that the solution to the equation is of the form

$$u(x, t) = \int_{-\infty}^t dt' \int_{-\infty}^{\infty} dx' G_d(t - t', x - x') i_{ext}(t', x') \quad (3.48)$$

where G_d is a Gaussian of the form

$$G_d(x, t) = \frac{1}{\sqrt{2\pi\sigma(t)}} \exp - \frac{x^2}{2\sigma^2(t)}. \quad (3.49)$$

Determine $\sigma(t)$ and discuss the result.

b) Use the method of mirror charges to discuss how the solution changes if the cable is semi-infinite and extends from zero to infinity.

c) Take the integral over space of the elementary solution of the non-leaky cable equation and show that the value of the integral does not change over time. Give an interpretation of this result.

d) Take the integral over space of the elementary solution of the normal leaky cable equation of a passive dendrite and derive an expression for its temporal evolution. Give an interpretation of your result.

5. Conduction velocity in unmyelinated axons

a) Using the simplified ion channel dynamics of Eq. (3.39), transform x and t to dimensionless variables using effective time and electrotonic constants.

b) A traveling pulse solution will have the form $u(x, t) = \tilde{u}(x - vt)$ where v is the conduction velocity. Find the ordinary differential equation that rules $\tilde{u}$.

c) Show that $\tilde{u}(y) = \frac{1}{1 + \exp(y)}$ with traveling speed $v = \frac{1-2a}{\sqrt{2}}$ is a solution.